

Logic Transformations

Commutativity of operators	$A \text{ and } B = B \text{ and } A$ $A \text{ or } B = B \text{ or } A$
Transformation of implication	$A \rightarrow B = \text{not}(A) \text{ or } B$ $\text{not}(A) \text{ or } B = A \rightarrow B$
Double negation	$\text{not}(\text{not}(A)) = A$ $A = \text{not}(\text{not}(A))$
Inverse Relation $p' = \text{inv}(p)$	$p(A,B) \rightarrow p'(B,A)$
Symmetric relation p Asymmetric relation p	$p(A,B) \rightarrow p(B,A)$ $p(A,B) \rightarrow \text{not}(p(B,A))$
Predicate implication $p \rightarrow q$ Not implication $p \rightarrow \text{not}(q)$	$p(A,B) \rightarrow q(A,B)$ $p(A,B) \rightarrow \text{not}(q(A,B))$
Transitivity	$A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C$
Contraposition	$A \rightarrow B = \text{not}(B) \rightarrow \text{not}(A)$
Tense (past tense vs present tense)	
De morgan's law	$\text{not}(A \text{ and } B) = \text{not}(A) \text{ or } \text{not}(B)$ $\text{not}(A \text{ or } B) = \text{not}(A) \text{ and } \text{not}(B)$